

The University of Texas at Austin
IN-CLASS WORK 6

M339D Intro to Financial Math

September 21, 2026

HEDGING WITH EUROPEAN CALL OPTIONS

PROBLEM 6.1: Sample FM(DM) #13.

Suppose that you short one share of a stock index for 50, and that you also buy a 60 –strike European call option that expires in 2 years for 10. Assume the effective annual interest rate is 3%. If the stock index increases to 75 after 2 years, what is the profit on your combined position?

SOLUTION: *The payoff is simply:*

$$-75 + (75 - 60)_+ = -60.$$

In words, as a short seller, you have to purchase the stock index back and you are going to take advantage of owning the call option on that index (as opposed to paying the higher market price). The initial cost is $-50 + 10 = -40$. So, the value of this initial cost in 2 years equals $-40 \cdot (1.03)^2 = -42.436$. The profit is

$$-60 + 42.436 = -17.564.$$

PROBLEM 6.2: *The “Very tasty goat cheese Co” sells artisan goat cheese at \$10 per oz. They need to buy 200 gallons of goat milk in six months to make 200 oz of their specialty fall-equinox cheese. Non-goat milk aggregate costs total \$500. They decide to buy six-month, \$5-strike call options on gallons of goat milk for 0.50 per call option.*

The continuously compounded, risk-free interest rate equals 0.04.

In six months, the price of goat milk equals \$6 per gallon. What is the profit of the company’s hedged position?

SOLUTION :

$$200 \times 10 - 200 \times 5 - 500 - 200 \times 0.50e^{0.02} = 397.98$$

PROBLEM 6.3: *The primary ingredient for a certain jeweler is gold which she intends to buy in exactly one year. She considers all of her other production-related expenses to be negligible.*

The jeweler uses exactly one ounce of gold to produce every one of her pieces, and will be able to sell every piece for \$1,000.

The jeweler models the market price of gold in one year as follows:

Gold price in one year	Probability
750 per ounce	0.2
850 per ounce	0.5
950 per ounce	0.3

The jeweler hedges the price of gold by buying a 1 – year call option with an exercise price of \$900 per ounce. The option costs \$10 per ounce now.

The continuously compounded risk-free interest rate is 5%.

Calculate the expected profit of the **hedged** portfolio per piece of jewelry produced.

SOLUTION: With $S(T)$ denoting the market price of gold at time $T = 1$, the jeweler's **hedged** profit per piece of jewelry can be expressed as

$$1,000 - \min(S(T), 900) - 10e^{0.05} = 989.4873 - \min(S(T), 900).$$

So, her expected **hedged** profit equals

$$989.4873 - \mathbb{E}[\min(S(T), 900)] = 989.4873 - (0.2 \times 750 + 0.5 \times 850 + 0.3 \times 900) = 144.4873.$$

PROBLEM 6.4: Suppose that you own one share of a stock index, currently priced at \$50, and that you write a \$60-strike European call option on it that expires in 2 years for \$10. Assume the effective annual interest rate is 3%.

You model the stock index price in 2 years as follows:

Stock index price in 2 years	Probability
40	0.25
65	0.50
90	0.25

Calculate the expected **payoff** of your **covered call** position.

SOLUTION: With $S(T)$ denoting the stock index price at time $T = 2$, the covered call payoff is

$$S(T) - (S(T) - 60)_+ = \min(S(T), 60).$$

Evaluating at each scenario: $\min(40, 60) = 40$, $\min(65, 60) = 60$, $\min(90, 60) = 60$. So,

$$\mathbb{E}[\min(S(T), 60)] = 0.25 \times 40 + 0.50 \times 60 + 0.25 \times 60 = 55.$$

REMARK 6.5: Note the premium of 10 and the interest rate of 3% are not needed here, since payoff (unlike profit) is not net of financing cost.

PROBLEM 6.6: A silver dealer owns exactly one ounce of silver, which — being non-perishable — she could sell at any time, but intends to sell in exactly one year. She considers all other costs negligible.

She models the market price of silver in one year as follows:

Silver price in one year	Probability
18 per ounce	0.5
22 per ounce	0.3
26 per ounce	0.2

To generate extra income, the dealer writes a 1-year covered call on her silver with strike price \$20 per ounce, and receives \$1.20 per ounce now for selling the option.

The continuously compounded, risk-free interest rate is 4%.

Calculate the expected profit of her **covered call** portfolio per ounce.

SOLUTION: With $S(T)$ denoting the market price of silver at time $T = 1$, the dealer's covered-call profit per ounce can be expressed as

$$\min(S(T), 20) + 1.20e^{0.04} = \min(S(T), 20) + 1.2490.$$

So her expected profit equals

$$\mathbb{E}[\min(S(T), 20)] + 1.2490 = (0.5 \times 18 + 0.3 \times 20 + 0.2 \times 20) + 1.2490 = 20.2490.$$

REMARK 6.7: Why would the dealer write this call rather than simply hold the silver unhedged? The median outcome in her distribution is \$18, below the \$20 strike, so the price she thinks **most likely** is one where the call expires worthless and she keeps the full premium — even though the mean, \$20.80, sits just above the strike because of the less-likely but higher \$26 outcome pulling it up. Since she already owns the ounce and is selling it in a year regardless, writing the call is not a directional bet; it monetizes her view that a rise much above \$20 is the less probable outcome. The \$1.20 premium lowers her effective cost basis and cushions a price decline, at the cost of capping her proceeds if silver does rally past \$20 — the usual trade-off of a covered call: income generation and partial downside protection, paid for by giving up the upper tail.